Color Space Fusion for Semantic Segmentation in Indian Adverse Weather Driving Scenes

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***Abstract*—Performing semantic segmentation in bad weather conditions is still extremely difficult because RGB sensors often have poor visibility in such conditions. In this study, we have explored various colour spaces to combine RGB and Near-Infrared images to improve the perception of environment in bad weather conditions. We assessed the YUV, LAB and HSV colour spaces to combine RGB and NIR images along with an RGB baseline for comparison. The IDD-AW dataset was used for this fusion across the YOLOv8 and YOLOv11 nano and small segmentation models. According to our experimental results, LAB fusion along with YOLOv11 yields the best results which had a SmIoU of 17.96 and a mIoU of 26.46. These results are a 10% improvement over the RGB baseline. Since YUV and LAB color spaces have separate luminance channels, they have higher scores when compared to HSV, which shows moderate improvements. This tells us that choosing the right color for space for the RGB and NIR fusion has a huge positive impact on semantic segmentation in conditions with fog, low light, rain and snow even for smaller models focusing on real-time inference.**

***Index Terms*—RGB-NIR fusion, semantic segmentation, YOLO, multi-spectral imaging, color space transformation, autonomous driving**

I. Introduction

Autonomous vehicles (AV) need a robust perception of their surrounding environments in various conditions of weather and lighting to drive with safety [1]. RGB cameras are widely used in AVs for spatial understanding but their performance degrades significantly under adverse conditions such as fog, rain, snow and low-light [2]. These limitations lead to safety risks for AVs that need to maintain consistent performance for safety purposes regardless of environmental conditions.

Fig. 1. RGB/NIR/Annotation example for rain, lowlight, fog and snow from top to bottom.

Near-infrared (NIR) radiation (around 700-1000 nanometer wavelength) is a part of the electromagnetic spectrum just outside the visible range [3]. Thus, NIR imaging offers information apart from RGB, that can be used to enhance

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perception by sensors in challenging scenarios [4]. However, effective integration of RGB and NIR information requires careful colour space representations and fusion techniques as incorrect transformations can lead to loss of information. Figure 1 shows an example of RGB, NIR and the segmented mask of different road scenes in rain, lowlight, fog and snow.

Recently, computer vision tasks like object detection and semantic segmentation have been strongly influenced by the YOLO (You Only Look Once) models [5], [6]. Their architectures have evolved significantly by incorporating different techniques like anchor-free detection, different backbones and feature aggregation [7]. YOLO variants are used not only in driving systems but also in diverse domains including medical imaging, video surveillance, agriculture and robotics.

This paper addresses the gap between multi spectral fusion techniques and modern segmentation architectures by evaluating color spaces for RGB-NIR integration using YOLO models. Our contributions include:
(1) Implementation and evaluation of four distinct color space fusion methods with mathematically defined transformations.
(2) Comprehensive performance analysis across four YOLO variants on adverse weather driving scenes.

II. Related Work

A. Semantic Segmentation in Autonomous Vehicles

Semantic segmentation is a computer-vision task that assigns labels to individual pixels in the image to mark the specific shapes of different objects and regions in the image. In AVs, semantic segmentation leads to a detailed perception of complex road scenes. This is helpful to navigate on roads by detecting obstructions and drivable road. Several datasets have been compiled to train and test segmentation methods specific to autonomous driving scenes. Annotated images of various road elements in urban settings are used to assess scene segmentation using the Cityscapes dataset [8]. IDD, which addresses the complex and unstructured traffic situations common on Indian roads, is another noteworthy dataset [9]. An extension of this dataset, IDD-AW, draws attention to adverse weather conditions with RGB-NIR image pairs [10]. Another dataset called the Berkeley Deep Drive (BDD100K), consists of a broad range of driving scenarios in varied environments [11]. A dataset called ACDC is made for segmentation in adverse weather scenarios such as rain and fog [12].

Some datasets, namely RGB-Thermal dataset [13] and RASMD (RGB and SWIR Multispectral Driving) dataset [14], capture complementary spectral information. This can enhance segmentation in adverse lighting or adverse weather conditions. Additionally, large-scale datasets like the Waymo Open Dataset [15], nuScenes [16] and Lyft Level 5 [17] have multi-modal sensor data which includes LiDAR, radar and cameras. In complex real-world driving scenarios, they support a variety of segmentation and perception tasks.

Recent methods leverage multi scale attention mechanisms and fusion techniques to exploit relevant features from these datasets which improves segmentation robustness in occluded and varying object scale scenarios [18]. Artificial datasets generated by driving simulators like CARLA, are utilized to augment real-world datasets to improve model training with diversity in scenes and conditions. [19].

B. Color Space Fusion

Color space transformations is a pre-processing step to convert an image from one space to another for processing and efficiency reasons. Transformations are also be used for image fusion which integrates additional information from different spaces into a single space. Commonly used color spaces such as HSV, LAB and YUV, separate luminance from chrominance channels. This is especially useful when incorporating near-infrared (NIR) data into RGB images [20], [21].

By giving each channel an adaptive weight, weighted fusion of spaces makes use of this separation. This improves the representation of features that are pertinent to the driving environment [22]. The quality of the fused image and the segmentation performance are both greatly impacted by the optimized tuning of these weights.

III. Methodology

A. Color Space Fusion

For RGB-NIR integration, we use four distinct color spaces through defined transformations. Below is a detailed description of the implementation.

After normalizing and flattening the input to a channel that matches the scale and format of RGB channels, the NIR images are utilized. This makes sure that the transformations and the pixel-wise operations are compatible.

1) YUV: 1) YUV: For enhancement of luminance with NIR information as well as retaining color characteristics with the chrominance elements, the luminance (Y) and the chrominance (U, V) components are separated by the YUV color space. The transformation employs ITU-R BT.601 standard coefficients as follows:

$$Y_{new} = 0.299R_{old} + 0.587G_{old} + 0.114B_{old} \tag{1}$$
$$U_{new} = -0.147R_{old} - 0.289G_{old} + 0.436B_{old} \tag{2}$$
$$V_{new} = 0.615R_{old} - 0.515G_{old} - 0.100B_{old} \tag{3}$$

NIR integration happens through weighted fusion on the Y channel:

$$Y_{fused} = \alpha \cdot Y_{new} + (1 - \alpha) \cdot I_{NIR} \tag{4}$$

where $\alpha = 0.7$ provides optimal balance between RGB colour preservation and NIR enhancement. RGB reconstruction follows inverse YUV transformation:

$$R_{new} = Y_{fused} + 1.139V_{fused} \tag{5}$$
$$G_{new} = Y_{fused} - 0.395U_{fused} - 0.581V_{fused} \tag{6}$$
$$B_{new} = Y_{fused} + 2.032U_{fused} \tag{7}$$

2) HSV: HSV color space allows manipulating intensity (V channel) while preserving hue and saturation information (H and S channels). RGB images were converted into HSV using existing OpenCV methods.

$$V_{fused} = \alpha \cdot V_{original} + (1 - \alpha) \cdot I_{NIR} \tag{8}$$

with $\alpha = 0.7$. The fused HSV is converted back to RGB using OpenCV methods with range clipping for valid color values.

3) LAB: CIE LAB color space provides uniform representation with separated lightness (L) and chromaticity (A,B) components. RGB images were converted into LAB using existing skimage methods.

$$L_{fused} = \alpha \cdot L_{original} + (1 - \alpha) \cdot I_{NIR}^{norm} \tag{9}$$

where $I_{NIR}^{norm} = (I_{NIR}/255) \times 100$ normalizes NIR values to LAB L channel range [0,100], and $\alpha = 0.8$.

4) Direct RGB Baseline: The baseline method uses unmodified RGB images to check performance benchmarks for fusion evaluation that allows for a quantitative assessment of multi spectral integration benefits.

B. YOLO Architecture Configuration

We evaluate four edge-deployable YOLO variants to represent different computational complexity and accuracy trade-offs.

TABLE I
YOLOV8 AND YOLOV11 SEGMENTATION MODEL COMPARISON

Model	CPU Speed (ms)	Params (M)	FLOPs (B)
YOLOv8n-seg	96.1	3.4	12.6
YOLOv8s-seg	155.7	11.8	42.6
YOLOv11n-seg	65.9	2.9	10.4
YOLOv11s-seg	117.6	10.1	35.5

All models are configured for segmentation with 30 class outputs corresponding to driving scene categories as per the IDD-AW dataset.

C. Training Configuration

Models are trained using identical hyperparameters to ensure fair comparison: - Image resolution: 640×640 pixels - Batch size: 32 (across 2 T4 GPUs) - Optimizer: AdamW with lr=0.000294, momentum=0.9. - Training epochs: 50 - Deterministic = True - Seed = 0

IV. Experimental Setup

A. Dataset and Preprocessing

IDD-AW [10] provides 5000 RGB-NIR image pairs with pixel-level annotation masks across four weather conditions: rain (1500 pairs), fog (1500 pairs), lowlight (1000 pairs) and snow (1000 pairs). 2-CMOS cameras with multi spectral prism were used to capture the images in RGB (400-670 nanometers) and NIR (740-1000 nanometers) spaces simultaneously.

B. Evaluation Metrics

Performance of semantic segmentation models on the IDD-AW dataset is assessed using four principal metrics: mAP_{50}, $\text{mAP}_{50\text{-}95}$, mean Intersection-over-Union (mIoU) and Safe mIoU (SmIoU), following recent evaluation protocols in robust driving scene segmentation [10].

1) Mean Intersection over Union (mIoU): Mean Intersection over Union (mIoU) is the standard metric for semantic segmentation, averaging the IoU across all classes:

$$\text{mIoU} = \frac{1}{N} \sum_{i=1}^{N} \frac{TP_i}{TP_i + FP_i + FN_i}$$

where TP_i, FP_i, and FN_i are the true positive, false positive, and false negative counts for class i and N is the total number of classes.

2) Safe mIoU (SmIoU): The SmIoU metric includes a penalty based on the semantic hierarchy and importance of classes especially for safety-critical categories.

First, we compute the intersection over union for pairs of predicted and ground truth classes:

$$I_{\text{safe}_{i,j}} = \frac{|gt_i \cap pred_j|}{|gt_i \cup pred_j|} \qquad I_{i,i} = \frac{|gt_i \cap pred_i|}{|gt_i \cup pred_i|} \tag{10}$$

where gt_i is the set of pixels belonging to class i in the ground truth, and $pred_j$ is the set of pixels predicted as class j.

Next, for each class i, safe IoU I_{safe_i}, is calculated with hierarchical penalties based on class distances in the label tree:

$$I_{\text{safe}_i} = \begin{cases} I_{i,i} - \sum\limits_{j \in C, j \neq i} \dfrac{d(i,j)}{n} I_{\text{safe}_{i,j}} & \text{if } i \in C_{\text{imp}} \\ I_{i,i} - \sum\limits_{j \in C_{\text{imp}}} \dfrac{d(i,j)}{n} I_{\text{safe}_{i,j}} & \text{otherwise} \end{cases} \tag{11}$$

Here, C is the set of all classes, C_{imp} is the subset of safety-critical classes, $d(i,j)$ is the tree distance between classes i and j in the hierarchy, and n is the number of hierarchy levels. Safe mIoU is the average of these weighted class-wise IoUs:

$$\text{SmIoU} = \frac{\sum_{i \in C} I_{\text{safe}_i}}{|C|} \tag{12}$$

V. Results and Analysis

A. Overall Performance Comparison

Figure 3 presents qualitative results of color space transformations across YUV, HSV and LAB spaces for fog, lowlight, rain and snow. Each row within every block corresponds to RGB, NIR, fused and difference images respectively. While the fused images often appear visually similar to the original RGB images, the difference visualizations highlight distinct pixel-level differences between fused and RGB images indicating changes across color spaces. Table II reports the aggregated performance of all fusion methods across the evaluated YOLO models. The results demonstrate that fusions consistently improves segmentation accuracy compared to the RGB baseline. Among the color spaces, LAB with YOLOv11s-seg achieves the highest overall performance with mIoU of

26.46 and SmIoU of 17.96, outperforming the corresponding RGB baseline (24.08/16.51). YUV fusion also shows competitive results across models, with YOLOv8s-seg attaining 26.13/18.33. HSV fusion provides moderate improvements over RGB but is generally less effective than YUV and LAB. Overall, LAB emerges as the most effective fusion strategy with YUV providing a strong balance of accuracy and consistency across models.

TABLE II
PERFORMANCE COMPARISON ACROSS FUSION METHODS AND YOLO ARCHITECTURES

Model	Colour Space	mIoU	SmIoU
YOLOv8n-seg	RGB	18.53	12.43
	YUV	19.39	12.96
	HSV	18.77	12.51
	LAB	19.81	13.24
YOLOv8s-seg	RGB	23.76	16.40
	YUV	26.13	18.33
	HSV	25.74	17.02
	LAB	26.45	18.41
YOLOv11n-seg	RGB	21.15	14.38
	YUV	22.81	16.22
	HSV	21.65	15.49
	LAB	23.44	16.83
YOLOv11s-seg	RGB	24.08	16.51
	YUV	25.11	17.58
	HSV	24.16	16.72
	LAB	26.46	17.96

B. Weather-Condition Analysis

Table III demonstrates that fusion-based approaches consistently outperform the RGB baseline across all YOLO architectures and weather conditions. The most substantial improvements are observed under fog where LAB fusion achieves the highest scores particularly with YOLOv8s-seg (30.95/22.28) and YOLOv11s-seg (30.96/21.73). These outcomes demonstrate the usefulness of NIR integration in low-visibility scenarios. In low-light conditions, both LAB and YUV spaces exhibit significant improvements as well, with YOLOv8s-seg (25.13/17.12) and YOLOv11s-seg (25.14/16.70). This shows that they clearly outperform their RGB counterparts. In snow, performance improvements are more moderate but consistent. For example with YOLOv11n-seg with LAB fusion (19.92/13.46) outperformed RGB (17.98/11.50). YOLOv11s-seg LAB (27.25/19.04) when compared to RGB (24.80/17.50) indicates that for rain, improvements were smaller but still consistent. All things considered, the LAB colour space comes out as the most effective fusion strategy across conditions, with YUV coming in second because of their separate light channels.

VI. CONCLUSION AND FUTURE WORK

A thorough assessment of color space fusion techniques for RGB-NIR integration in inclement weather conditions was provided in this work. According to the experimental results on the IDD-AW dataset, LAB fusion achieves the most superior segmentation performance in fog and low-light scenarios specifically. Across all methods, YOLOv11 architectures perform better than their YOLOv8 counterparts, illustrating the advantages of architectural enhancements for robust multimodal perception.

TABLE III
PERFORMANCE COMPARISON ACROSS FUSION METHODS AND YOLO ARCHITECTURES BY WEATHER CONDITIONS

Model	Colour Space	Lowlight		Snow		Rain		Fog	
		mIoU	SmIoU	mIoU	SmIoU	mIoU	SmIoU	mIoU	SmIoU
YOLOv8n-seg	RGB	17.60	11.56	15.75	9.94	19.09	13.18	21.68	15.04
	YUV	18.42	12.05	16.48	10.37	19.97	13.74	22.69	15.68
	HSV	17.83	11.63	15.95	10.01	19.33	13.26	21.97	15.14
	LAB	18.82	12.31	16.84	10.59	20.40	14.03	23.18	16.03
YOLOv8s-seg	RGB	22.57	15.25	20.20	13.12	24.47	17.38	27.80	19.85
	YUV	24.82	17.05	22.21	14.66	26.91	19.43	30.58	22.18
	HSV	24.45	15.83	21.88	13.62	26.51	18.04	30.12	20.59
	LAB	25.13	17.12	22.48	14.73	27.24	19.51	30.95	22.28
YOLOv11n-seg	RGB	20.09	13.37	17.98	11.50	21.78	15.24	24.75	17.41
	YUV	21.67	15.08	19.39	12.98	23.49	17.19	26.69	19.63
	HSV	20.57	14.41	18.40	12.39	22.30	16.42	25.33	18.74
	LAB	22.27	15.65	19.92	13.46	24.14	17.84	27.43	20.37
YOLOv11s-seg	RGB	22.88	15.35	20.47	13.21	24.80	17.50	28.17	19.98
	YUV	23.85	16.35	21.34	14.06	25.86	18.63	29.39	21.28
	HSV	22.95	15.55	20.54	13.38	24.88	17.72	28.27	20.23
	LAB	25.14	16.70	22.49	14.37	27.25	19.04	30.96	21.73

By directly extending segmentation models to accept NIR as a channel along with RGB, we intend to move beyond static color space transformations in future research. Instead of relying on fixed alpha parameters, this method would allow the network to learn optimal fusion weights during training. This design could have the ability to capture richer cross-modal representations that dynamically adapt to scene conditions, as well as enhance robustness across weather scenarios. Furthermore, when multi-spectral modalities like thermal infrared and radar are integrated in a similar fashion, it can improve the perception pipeline for autonomous driving systems.

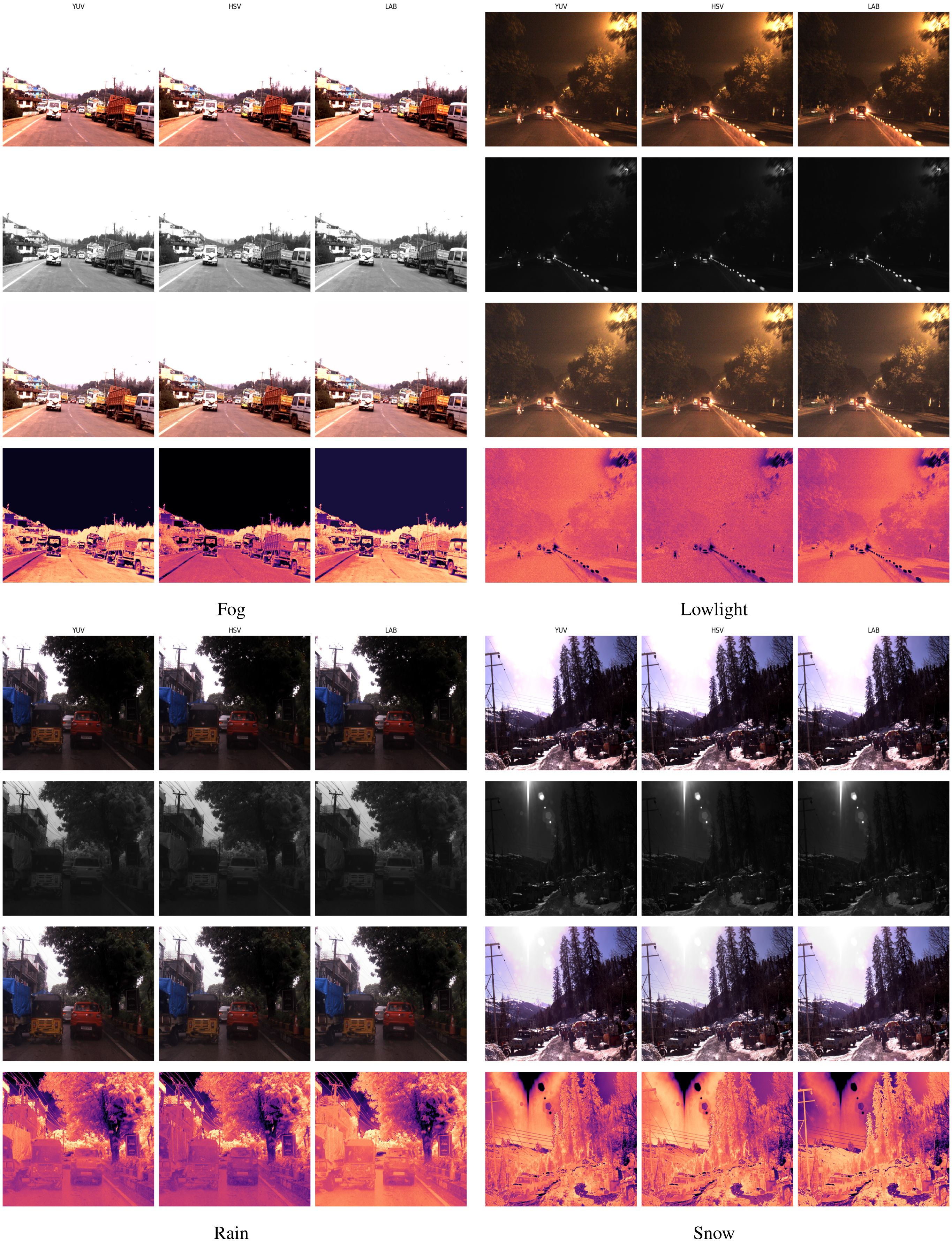

Fig. 2. RGB, NIR, Fused and Image Difference between fused and RGB image from top to bottom across YUV, HSV and LAB spaces from left to right.

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